HERITAGEAI: ARTIFICIAL INTELLIGENCE SYSTEM FOR AUTOMATIC IDENTIFICATION AND CLASSIFICATION OF CULTURAL PATTERNS

A MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **"HERITAGEAI: ARTIFICIAL INTELLIGENCE SYSTEM FOR AUTOMATIC IDENTIFICATION AND CLASSIFICATION OF CULTURAL PATTERNS"** is the bonafide work of **"JATHIN V BANJAN (USN: NN25MCA051)"** who carried out the project work under my supervision.

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ABSTRACT

India possesses a rich and diverse tapestry of indigenous art forms, folk paintings, and handloom textile traditions. Among these, Warli tribal art from Maharashtra, Madhubani folk paintings from Bihar, and traditional Indian textile motifs (incorporating Paisley and Bandhani patterns) reflect centuries of cultural heritage and traditional knowledge. However, rapid globalization, lack of formal digitization, and subjective manual cataloging present critical risks of cultural attrition, misattribution, and loss of provenance.

This mini-project report presents **HeritageAI**, a research-grade, locally deployable artificial intelligence platform designed for the automatic identification, classification, explainability, and digital preservation of Indian cultural patterns. The core computer vision pipeline utilizes a customized Convolutional Neural Network (CNN) architecture engineered for three distinct pattern categories: Warli Art Motifs, Madhubani Paintings, and Indian Textile Patterns.

The image processing methodology enforces rigorous aspect-ratio preserving resizing with zero-padding (letterboxing) to 224 × 224 × 3 RGB and pixel intensity normalization to [0.0, 1.0]. The deep learning architecture incorporates stacked Conv2D feature extraction layers, 2 × 2 max-pooling, dense fully connected layers with ReLU activations, dropout regularization (rate 0.45) to prevent overfitting, and a three-unit Softmax classification layer trained via the Adam optimizer (learning rate 0.0001) under Sparse Categorical Crossentropy loss. To foster algorithmic transparency, the system implements Gradient-weighted Class Activation Mapping (Grad-CAM) to generate spatial attention heatmaps illustrating salient feature regions that influence model predictions.

The software architecture integrates a React 18 / Vite / TypeScript single-page application, an enterprise Java 21 / Spring Boot 3 RESTful microservice with JWT authentication and Flyway database migrations, and a Python 3.11 / FastAPI deep learning inference service supported by PostgreSQL. Experimental evaluations demonstrated high classification accuracy alongside low inference latency, validating the system as an effective preservation instrument for archivists, museum curators, and cultural researchers.

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CHAPTER 1
INTRODUCTION

Cultural heritage embodies the historical memory, spiritual philosophy, and aesthetic identity of human civilization. In the Indian subcontinent, traditional art and handloom forms have flourished over millennia, serving not merely as decorative expressions but as visual languages reflecting tribal folklore, sacred geometry, and socio-ecological connections. Prominent among these indigenous visual expressions are: (1) **Warli Art** from Maharashtra, characterized by rudimentary geometric shapes—circles representing celestial bodies, triangles depicting sacred mountains, and stylized human stick figures engaged in ritual Tarpa circle dances; (2) **Madhubani Paintings** from Bihar, renowned for prominent double outlines, fine parallel line hatchings, and sacred motifs; and (3) **Indian Textile Patterns**, encompassing Paisley (*Kalka*) motifs and Bandhani tie-dye grid dot matrices.

1.1 Problem Definition

Traditional cataloging of cultural heritage artifacts is predominantly manual, labor-intensive, and reliant upon the subjective expertise of specialized art historians. Archival institutions and museums often lack automated computational tools capable of accurately distinguishing between visual traditions sharing overlapping primitives, ingesting arbitrary digital photographs while mitigating distortions introduced by non-uniform aspect ratios, explaining deep neural network decisions, and decoupling AI predictions from verified institutional metadata.

The fundamental objective of this project is to develop and validate **HeritageAI**, an automated computer vision intelligence and digital preservation platform that performs multi-class CNN inference across Warli, Madhubani, and Indian Textile categories, provides Grad-CAM attention visualizations, and securely persists catalog records through a verified review workflow.

1.2 Objectives

The core objectives accomplished in this project are:

1. **Research-Grade Preprocessing**: Implement EXIF orientation correction, aspect-ratio preserving letterbox zero-padding to 224 × 224 × 3 RGB, and normalization to [0.0, 1.0].
2. **Custom CNN Architecture**: Stacked Conv2D blocks with ReLU, 2 × 2 max-pooling, Dense layers, 0.45 dropout, and Softmax output trained under Adam optimizer (lr = 0.0001).
3. **Model Interpretability (Grad-CAM)**: Generate spatial attention heatmaps illustrating salient feature regions.
4. **Enterprise Microservices**: Java 21 Spring Boot 3 REST API with JWT security and PostgreSQL persistence.
5. **Digital Archiving Workflow**: Provide curatorial review tools, low-confidence alerts (< 0.60), and CSV export.

1.3 Challenges

Key challenges addressed include aspect ratio distortion caused by arbitrary photograph shapes, high intra-class texture variations across mud walls and handloom fabrics, algorithmic interpretability for curatorial trust, and data integrity safeguards preventing automated AI predictions from overwriting

historical provenance records.

1.4 Literature Survey

Early cultural classification approaches relied on handcrafted feature descriptors such as SIFT and HOG, which struggled to generalize across textural substrate variations. General-purpose deep architectures (AlexNet, ResNet-50) demonstrated high accuracy but carried substantial parameter counts and higher latency. The research foundation paper validated that a tailored, stacked Conv2D architecture achieves optimal accuracy while remaining lightweight for local institutional deployment.

Architecture / Model	Mean Accuracy	Macro Precision	Macro Recall	Macro F1	Latency
Proposed HeritageCNN	**96.2%**	**0.958**	**0.946**	**0.952**	**0.22 s/img**
AlexNet Baseline	79.5%	0.781	0.764	0.772	0.12 s/img
ResNet-50 Baseline	86.8%	0.865	0.851	0.858	0.45 s/img

Table 1.1: Benchmark Comparison of Visual Classification Architectures

CHAPTER 2
SOFTWARE REQUIREMENT SPECIFICATION

2.1 Functional Requirements

1. **User Authentication & RBAC**: BCrypt password hashing, stateless JWT tokens, role gating (ADMIN, RESEARCHER, ARCHIVIST).
2. **Image Ingestion & Preprocessing**: Format validation (JPG, PNG, WebP), EXIF orientation transpose, aspect-ratio preserved letterbox resize to 224 × 224 × 3, and normalization to [0.0, 1.0].
3. **Pattern Classification Engine**: 3-class Softmax probability estimation and inference latency measurement.
4. **Low-Confidence Alert**: Flagging predictions with top confidence score under 0.60.
5. **Explainable AI (Grad-CAM)**: Extraction of final convolutional gradient maps overlaid on the original image.
6. **Preservation Catalog & Review Workflow**: Artifact metadata persistence and reviewer correction workflows.
7. **Data Export & Audit Trail**: CSV export of catalog records and automated audit logging.

2.2 Non-functional Requirements

The system maintains single-image inference latency under 300 ms on standard CPUs, ensures fault tolerance through decoupled microservices, secures user credentials via BCrypt, and obfuscates uploaded filenames using UUIDs to prevent directory traversal.

2.3 Hardware and Software Requirements

Component	Minimum Requirement	Recommended Specification
Processor	Intel Core i5 (4 Cores, 2.5 GHz)	Intel Core i7 / AMD Ryzen 7 (8 Cores)
Memory (RAM)	8 GB DDR4	16 GB / 32 GB DDR4/DDR5
Storage	10 GB SSD space	50 GB NVMe Solid State Drive
GPU (Optional)	CPU Inference Supported	NVIDIA RTX 3060 / T4 GPU (CUDA 12)

Table 2.1: Minimum Hardware Specifications

Layer / Component	Technology / Framework	Version	Role / Purpose
Backend Platform	Java Development Kit (JDK)	21 LTS	Core Enterprise Language
Backend Framework	Spring Boot (Security, JPA)	3.2.3	REST API & Business Logic
AI Microservice	Python / FastAPI	3.11 / 0.109	Asynchronous AI Service
Deep Learning	TensorFlow / Keras	2.15.0	CNN Training & Inference
Database Engine	PostgreSQL / Flyway	16 / 10.x	Relational Database Storage
Frontend Stack	React / Vite / TypeScript	18.2 / 5.1	Responsive Client SPA

Table 2.2: Software Specifications and Technologies

CHAPTER 3
SYSTEM DESIGN

3.1 High Level Design

HeritageAI employs a three-tier microservice architecture isolating compute-intensive deep learning tasks from transactional data management:

1. **Presentation Tier**: React 18 / TypeScript SPA managing interactive uploads, probability charts, and catalog browsing.
2. **Application Backend Tier**: Spring Boot 3 enterprise service validating JWT tokens, managing file storage, and enforcing business logic.
3. **AI Computation Tier**: Python FastAPI microservice executing preprocessing, CNN inference, and Grad-CAM generation.
4. **Data Tier**: PostgreSQL 16 maintaining normalized relational records and audit histories.

3.2 Detailed Design

Data Flow Sequence: (1) Client uploads cultural image via multipart form; (2) Spring Boot validates file type and stores it to disk with a UUID key; (3) Spring Boot calls Python service /predict endpoint; (4) Python service letterboxes image to 224x224x3, scales pixels to [0, 1], and runs HeritageCNN; (5) Grad-CAM generates attention heatmap; (6) AI service returns probabilities and latency; (7) Backend saves prediction event to PostgreSQL; (8) Frontend displays results with Save to Archive capability.

CHAPTER 4
SYSTEM IMPLEMENTATION

4.1 Module Description

1. **Preprocessing Module** (preprocessing.py): Enforces aspect-ratio preserving letterbox padding. Scaling scale = min(224/w, 224/h) centers the image within a black canvas, preventing motif distortion.
2. **HeritageCNN Neural Module** (model.py): Stacked Conv2D layers with 32, 64, and 128 filters, 2x2 max-pooling, 256-unit dense layer, 0.45 dropout regularization, and 3-class Softmax output.
3. **Grad-CAM Explainability Module** (gradcam.py): Computes gradients of the target class with respect to target_conv_layer activations, generating spatial heatmaps highlighting influential motif regions.
4. **Backend & Security Engine** (backend/): Spring Security JWT filter chain, entity repositories, and REST controllers.

Layer Group	Layer Type	Configuration	Output Shape	Activation
Input	InputLayer	RGB Channels	(224, 224, 3)	—
Block 1	Conv2D x 2 + MaxPool	32 filters, 3x3, Stride 1	(112, 112, 32)	ReLU
Block 2	Conv2D x 2 + MaxPool	64 filters, 3x3, Stride 1	(56, 56, 64)	ReLU
Block 3	Conv2D x 2 + MaxPool	128 filters, 3x3, Stride 1	(28, 28, 128)	ReLU (target_conv)
Dense	Flatten + Dense	256 Units	(256)	ReLU
Dropout	Dropout	Rate = 0.45	(256)	—
Output	Dense	3 Units (Warli, Madhubani, Textile)	(3)	Softmax

Table 4.1: Custom HeritageCNN Architectural Specification

4.2 Tools, Language and RDBMS used

The system is implemented using Java 21 LTS, Spring Boot 3.2.3, Python 3.11, FastAPI, TensorFlow/Keras 2.15, PostgreSQL 16, React 18, Vite 5, TypeScript 5, Tailwind CSS, and a self-contained Jupyter Research Notebook.

CHAPTER 5
TESTING

Rigorous verification was conducted across all microservices using Pytest for the AI engine, JUnit 5 for the backend, and integration testing for end-to-end classification.

Test ID	Module	Test Scenario	Expected Result	Status
TC-01	Preprocessing	Non-square image (800x400)	Letterbox padding to (224,224,3) in [0,1]	**PASS**
TC-02	Ingestion	Unsupported file (.bmp)	HTTP 400 Bad Request returned	**PASS**
TC-03	Inference	Valid cultural image	Probabilities sum to 1.0; latency < 0.3s	**PASS**
TC-04	Confidence	Ambiguous input (< 0.60)	low_confidence_warning set to true	**PASS**
TC-05	Grad-CAM	includeGradcam = true	Valid Base64 PNG heatmap returned	**PASS**
TC-06	Security	Admin login credentials	JWT Bearer token emitted	**PASS**
TC-07	Archive	Query category 'warli'	Filtered Warli subset returned	**PASS**
TC-08	Review	Curator correction	Original AI preserved; verified updated	**PASS**

Table 5.1: Test Cases and Execution Results for Core System Modules

CHAPTER 6
CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The **HeritageAI** platform successfully implements and operationalizes the research paper and patent application for automatic identification, classification, and digital preservation of Indian cultural patterns. By coupling an aspect-ratio preserving computer vision preprocessing pipeline with a tailored stacked Conv2D architecture, the system achieves research-grade classification across Warli, Madhubani, and Indian Textile motifs.

The integration of Grad-CAM explainability bridges the gap between deep learning and art historical curation by visually highlighting the spatial motif regions responsible for classification. Furthermore, the multi-tier architecture—combining a React/TypeScript interface, an enterprise Spring Boot REST API, a FastAPI AI microservice, and a PostgreSQL relational repository—ensures robust data provenance, prevents algorithmic overwriting of historical records, and provides cultural institutions with a turnkey preservation platform.

6.2 Future Enhancements

1. **Multi-Label Object Detection & Semantic Segmentation**: Extend the architecture using YOLOv8 or Mask R-CNN to localize and segment individual motifs within large-scale composite murals.
2. **Expansion of Cultural Pattern Taxonomy**: Broaden the classification taxonomy from three categories to encompass over twenty indigenous South Asian art traditions, including Kalamkari, Pattachitra, Phad, Gond, and Tanjore paintings.
3. **Edge Quantization for Offline Field Archiving**: Quantize trained Keras models into TensorFlow Lite (TFLite) and ONNX formats to enable offline mobile inference on portable devices used in rural field research.

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